

# Nimbus Advanced Agricultural Benchmarking

Sheerwater Benchmark · Method Note  
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**TL;DR. Actionable days beyond baseline:** for each grid cell, take the longest lead at which it still meets the actionable percent-good cutoff, subtract the same horizon for the baseline, then average over all cells in the selected country or countries.

## 1 Motivation

In the Nimbus program, our north star is that smallholder farmers across sub-Saharan Africa have the forecasts they need, backed by their national meteorological and hydrological services, to make critical agricultural decisions during the rainy season. Benchmarking must help us ask whether forecasts are **good enough** for those decisions: scores should look good when a farmer could have used the forecast well, and bad when it would have misled them.

Standard daily rainfall metrics (MAE, CRPS, brittle contingency scores) often fail that test: they mix wet and dry regimes, punish tiny timing shifts with double penalties, or hinge on hard thresholds that do not match farmer-relevant “wet enough” / “dry enough” questions. Sheerwater’s agricultural event metrics instead verify specific decision-relevant events with temporal softening so that rain a day or two off is not scored as a full miss.

### Agriculturally relevant events

In our first implementation, we focus on the following events. “Good enough” is defined separately for each event (magnitude, dryness, timing, or accumulation), and can be updated as application needs change.

| Event             | Agricultural decision                                                        | Definition                                                                                                     | How can we measure good enough?                                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extreme rain days | Flood / erosion risk; whether to delay field work or protect harvested crops | Days where 1-day rainfall is at or above the local 90th percentile of daily precip from IMERG Final, 1998–2025 | Score = SMAPE of rainfall in the 5-day window centered on the event ( $\pm 2$ days). <b>Good enough:</b> $\text{SMAPE} \leq 0.30$ (within about 30% of the true rainfall in that window) |

| Event                              | Agricultural decision                                                                            | Definition                                                                                                                                                                                                                                     | How can we measure good enough?                                                                                                                                 |
|------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In-season dry spells               | When to plant; whether to irrigate, replant, or expect moisture stress during crop establishment | After the season has started (first > 4 mm rain day near 10% of seasonal accumulation) and before 40% of seasonal rain has fallen (seasons with < 150 mm total rain excluded): any 10-day window with < 15 mm total rain                       | Score = forecast rainfall (mm) in that same 10-day window. <b>Good enough:</b> forecast rain $\leq 20$ mm (at most 20 mm predicted where observations were dry) |
| Early-season rainfall accumulation | When to plant; whether early-season moisture is on track for crop establishment                  | First day in the season when cumulative rainfall reaches 100 mm; the evaluated quantity is rainfall accumulated over the preceding 30 days (right-aligned). For forecast leads shorter than 30 days, earlier days are filled from observations | Score = MAE (mm) of that 30-day accumulation vs truth. <b>Good enough:</b> MAE $\leq 30$ mm (the forecasted 30-day total is within 30 mm of the observed total) |
| Start of season date*              | When to plant / prepare land for the rainy season                                                | First date in each season meeting a start-of-season rule (ICPAC, CHC, or Moron & Robertson)                                                                                                                                                    | Score = absolute error (days) between forecast and observed start-of-season date. Good-enough bar TBD                                                           |

\*Implementation in progress.

**Reading the good-enough bars.** For extreme rain days we use  $\text{SMAPE} = |F - O| / (|F| + |O|)$ .  $\text{SMAPE} \leq 0.30$  means the absolute error is at most 30% of the combined forecast and observed magnitude. In plain terms, the forecast rainfall in the softened window was within about 30% of the true rainfall. For dry spells,  $\leq 20$  mm means the forecast did not put more than 20 mm of rain into a 10-day window that was observed dry ( $< 15$  mm). For early-season accumulation, MAE  $\leq 30$  mm means the forecast's 30-day total missed the observed total by at most 30 mm.

## Observation-triggered and forecast-triggered events

For event metrics such as extreme rain days and dry spells, an event can be defined from the observations or from the forecast. We score both:

- **Observation-triggered:** events found in the observations. Poor scores here are *false negatives*: rain (or a dry spell) happened, but the forecast missed it.
- **Forecast-triggered:** events found in the forecast. Poor scores here are *false positives*: the forecast warned of an event that was not observed.

KR Tracking’s default “both” mode combines the two by taking the **worse** of the observation-triggered and forecast-triggered scores at each cell and lead (equivalently: the maximum error, or the minimum percent-good). That way a forecast cannot look good by catching observed events while over-warning, or by staying quiet while missing real events; both failure modes count.

The **KR Tracking** dashboard then summarizes those event-based scores as:

1. how many grid cells clear a chosen quality cutoff at each lead (**good cells at a cutoff**), and
2. how far each cell’s skill extends in lead time relative to a baseline (**days beyond baseline**).

This note defines those two quantities.

## 2 Setup

Let  $F$  be a candidate forecast and  $B$  a baseline forecast (e.g. climatology or a reference NWP product). For a chosen agricultural event metric, each of  $F$  and  $B$ , at lead day  $L$  and grid cell, has a **percent-good** score in  $[0, 1]$ : the fraction of verified events meeting the metric’s good-enough bar.

Fix two cutoffs on that score. These cutoffs are flexible and can be changed for different applications or risk tolerances; for now we use  $\theta_{\text{info}} = 0.50$  and  $\theta_{\text{act}} = 0.75$ :

| Cutoff      | Symbol                        | Meaning                                                                                                                                                                                         |
|-------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Informative | $\theta_{\text{info}} = 0.50$ | At least 50% of events at that cell and lead are well predicted. Useful signal (worth watching), but usually not enough to drive operational forecast alerts or firm farm decisions on its own. |
| Actionable  | $\theta_{\text{act}} = 0.75$  | At least 75% of events are well predicted. High enough consistency to support decision-grade use (e.g. alerts, irrigation timing, or planting guidance).                                        |

Leads considered for the horizon KPI are

$$L \in \{7, 14, 21, 28\},$$

i.e. forecast weeks 1–4. We do not evaluate weeks 5 and 6 ( $L = 35, 42$ ) here, because skill at those leads was consistently very low. A cell **passes** cutoff  $\theta$  at lead  $L$  when its percent-good is at least  $\theta$ .

### Probabilistic forecasts

Ensemble forecasts use the same event definitions and good-enough bars as deterministic forecasts. We do **not** collapse the ensemble to a single mean field before scoring. Instead, each ensemble member is scored with the deterministic metric and marked good or bad against the good-enough bar; the resulting good-enough percentage is then averaged across all ensemble members. KR Tracking can be run in deterministic or probabilistic mode (separate cached tables). In probabilistic

mode, cell curves and days-beyond KPIs use this member-averaged rate: how often a draw from the ensemble would have been good enough, not whether the ensemble-mean rainfall field alone cleared the bar.

### 3 Good cells at a cutoff

At a fixed lead  $L$ , define

$$G_F(L, \theta) = \#\{\text{cells with percent-good} \geq \theta\}.$$

The dashboard’s threshold curve plots  $G_F(L, \theta)$  versus  $\theta$ . The Informative and Actionable markers are narrow guides at  $\theta_{\text{info}}$  and  $\theta_{\text{act}}$ . The lead-panel deltas are

$$\Delta G(L, \theta) = G_F(L, \theta) - G_B(L, \theta),$$

reported as a **number of cells** versus baseline  $B$ .

Optional table views also report  $G_F(L, \theta_{\text{info}})$  and  $G_F(L, \theta_{\text{act}})$  directly (“Cells @ Informative / Actionable”). A separate “Expected Cells” view is an AUC-like summary of the threshold curve: the average of  $G_F(L, \theta)$  over all cutoffs  $\theta \in [0, 1]$ . It is a curve-wide score, not the days-beyond KPI.

### 4 Days beyond baseline

Days beyond baseline answers: *relative to the baseline, how much farther in lead time does this forecast stay good enough?*

#### Per cell

For each grid cell  $c$  and cutoff  $\theta \in \{\theta_{\text{info}}, \theta_{\text{act}}\}$ , define the **horizon** as the longest lead at which the cell still passes:

$$H_F(c, \theta) = \max\{L \in \{7, 14, 21, 28\} : \text{percent-good}_F(c, L) \geq \theta\}, \quad (1)$$

or 0 if the cell never clears  $\theta$  at those leads. A null percent-good at a lead is a noop: that lead is dropped for both  $F$  and  $B$  on the cell (horizons use only shared usable leads). It is not treated as a fail.

The cell’s days beyond baseline are

$$D_F(c, \theta) = H_F(c, \theta) - H_B(c, \theta), \quad (2)$$

with both horizons computed on the intersection of leads where  $F$  and  $B$  have usable percent-good. Cells with no shared usable leads are omitted from the mean.

## Region summary

Regions are selectable as a single country or multiple countries. The reported value is the **mean over all grid cells** in the selected country or countries:

$$\text{Days}(F, B; \theta) = \frac{1}{n} \sum_c D_F(c, \theta), \quad (3)$$

where the sum runs over the  $n$  comparable cells in that selection. Separate values are shown for Informative ( $\theta_{\text{info}}$ ) and Actionable ( $\theta_{\text{act}}$ ). The baseline column is always 0. Forecasts with no overlapping usable cells show as missing, not as zero (so short-range or sparse models are not scored as “tied with baseline”).

Against a baseline with horizon 0 everywhere, that mean is easy to read: every cell actionable out to lead 28 gives +28; every cell actionable only at lead 7 gives +7; if 3 of 15 cells are actionable out to lead 14 and the rest are unchanged, the region mean is  $3 \times 14/15 = +2.8$ ; and matching the baseline everywhere gives 0. A cell-level +14 means that cell clears the cutoff two weeks farther than the baseline on that same cell.

## 5 Limitations and next steps

- **Training / testing conflict.** Several ML forecasts were trained on years that overlap the evaluation period (e.g. training through 2021), so KR Tracking scores can be optimistic relative to a true out-of-sample holdout. *Planned:* a rolling overfit sensitivity study that varies the evaluation window relative to each model’s training cutoff.
- **Probabilistic evaluation.** Ensemble forecasts are scored per member and then averaged, which captures **expected** error across the ensemble rather than the true error of a single realized forecast or a proper score of the full distribution. *Planned:* compute a median event from the ensemble, then evaluate that deterministic event.
- **Equal cell weighting.** Regional means treat every grid cell equally, so empty or sparsely farmed cells weigh as much as intensive agricultural areas. *Planned:* weight cells by arable land or farming population.
- **Discrete weekly leads.** Horizons are evaluated only at four leads ( $L \in \{7, 14, 21, 28\}$ : weeks 1–4). Sub-weekly skill is ignored: a cell that remains good enough out to day 10 is still tracked as horizon 7. *Planned:* denser lead sampling (e.g. daily or every few days) so mid-week gains are counted.

Implemented in the KR Tracking dashboard.